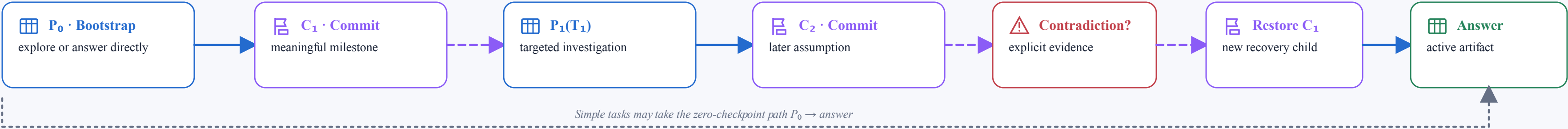


Phase Reasoning, Semantic Checkpoints, and Recovery

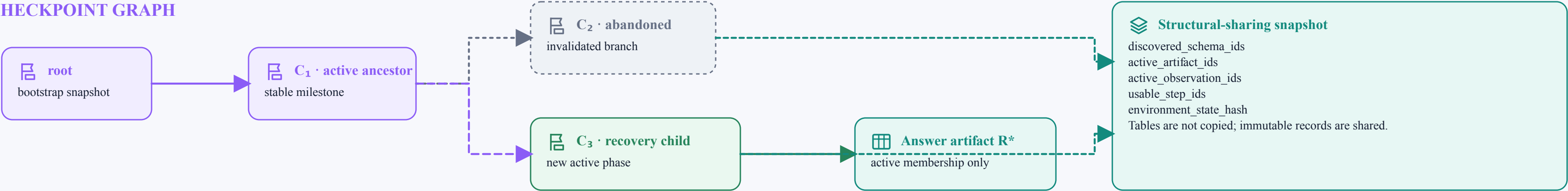
Checkpoints control active working memory; they do not create database evidence

DIAGNOSTIC ONLY

PHASE TIMELINE



CHECKPOINT GRAPH



COMMIT CONTRACT

progress_summary · 1–5 verified milestone statements
remaining_uncertainties · 0–5 unresolved semantic questions
next_targets · 1–3 objectives for the next phase
Commit only after answer-relevant semantic progress.

RESTORE CONTRACT

checkpoint_id · an available snapshot or root
reason · contradiction + invalidated assumption + why this node
next_targets · 1–3 recovery objectives
Old records remain immutable; inactive handles cannot be referenced.

PHASE TRANSCRIPT RESET

After commit or restore: clear the old provider phase transcript.
Retain compact CHECKPOINT HISTORY and current EnvironmentState.
Carry only grounded producer identity across phases.
Provider-private reasoning is archived for audit, not re-injected as fact.

F08 · Checkpoint snapshot and logical hash

$$C_k = \langle D_k, A_k, O_k, U_k, h_k \rangle, \quad h_k = \text{SHA256}(\text{canon}(S_k))$$

F09 · Exact restore and active path

$$\text{Restore}(C_j) : \quad S \leftarrow S(C_j), \quad \mathcal{P}_{\text{active}} \leftarrow \text{Ancestors}(C_j) \cup \{C_{\text{recovery}}\}$$

Global identifiers are monotonic and never reused, including after restore.